

Discontinuation of Renin-Angiotensin System Inhibitors during Acute Kidney Injury Episode and All-Cause Mortality

Target Trial Emulation Studies

Sheng Nie , Yanqin Li, Yinfang Sun, Shan Chen, Xian Shao , Mingzhen Pang, Shiyu Zhou, Licong Su , Ruixuan Chen , Fan Luo, Xin Xu , and Fan Fan Hou , on behalf of the CRDS Study Investigators*

Key Points

- It is uncertain whether the discontinuation of renin-angiotensin system (RAS) improves prognosis among patients during an episode of hospital-acquired AKI.
- Discontinuation of RAS inhibitors within 2 days after AKI was associated with lower risk of all-cause mortality as compared with continuation.
- Temporary discontinuation of RAS inhibitors during AKI episodes may improve outcomes in long-term users.

Abstract

Background Renin-angiotensin system inhibitors (RASi) are widely used among patients with cardiovascular disease, diabetes, and CKD, which are high-risk populations for AKI. Evidence for the optimal management of RASi during an episode of AKI is lacking.

Methods We conducted a multidatabase cohort study using sequential target trial emulation framework from the China Renal Data System and Medical Information Mart for Intensive Care IV (MIMIC-IV) databases. Adult patients with hospital-acquired AKI who had been receiving angiotensin-converting enzyme inhibitors or angiotensin receptor blockers treatment for at least 90 days were included. The primary outcomes were 30- and 180-day all-cause mortality. Adjusted cumulative incidences and risk differences were estimated using weighted pooled logistic regression models.

Results A total of 27,003 person-trials were identified from the China Renal Data System database (median [interquartile range] age, 70 [58–78] years; 12,972 [60%] male) and the MIMIC-IV database (median [interquartile range] age, 73 [63–82] years; 2825 [53%] male). The adjusted cumulative incidence of 30-day all-cause mortality was lower among person-trials with RASi discontinuation within 2 days after the hospital-acquired AKI than continuation (4.36% versus 5.91%), with a risk difference of -1.55% (95% confidence interval, -2.43% to -0.55%). Discontinuation of RASi was consistently associated with lower risk of all-cause mortality in the MIMIC-IV database, with similar beneficial associations observed across stratified analyses and multiple sensitivity analyses.

Conclusions In this study, stopping RASi within the first 2 days of AKI detection was associated with a lower risk of all-cause mortality.

JASN 00: 1–11, 2025. doi: <https://doi.org/10.1681/ASN.0000000775>

Division of Nephrology, State Key Laboratory of Multi-Organ Injury Prevention and Treatment, National Clinical Research Center for Kidney Disease, Nanfang Hospital, Southern Medical University, Guangzhou, China

Correspondence: Prof. Fan Fan Hou or Dr. Xin Xu, email: ffhouguangzhou@163.com or xux007@163.com

Received: February 16, 2025 **Accepted:** June 13, 2025

Published Online Ahead of Print: June 18, 2025

S.N., Y.L., and Y.S. contributed equally to this work.

*CRDS Study Investigators: Xin Xu, Sheng Nie, Hong Xu, Bicheng Liu, Jianping Weng, Chen Chunbo, Huafeng Liu, Qiongqiong Yang, Hua Li, Yaozhong Kong, Guisen Li, Qijun Wan, Yan Zha, Ying Hu, Gang Xu, Yongjun Shi, Yilun Zhou, Guobin Su, Ying Tang, Jie Shen, Chen Wang, Dehui Liu, Yihua Bai, Pengli Luo, Wenli Chen, Jianghua Chen, Chen Lu, Siyuan Teng, Yang Li, and Mengchun Gong.

Introduction

AKI is a common and life-threatening complication in hospitalized patients, especially in those with cardiovascular disease, diabetes mellitus, and CKD.¹ It has been consistently associated with higher short-term and long-term risks of mortality. Currently, there are no proven effective therapies for AKI, and the optimal management strategies for improving outcomes during or after an episode of AKI remain poorly defined.

Renin-angiotensin system inhibitors (RASi), particularly angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin receptor blockers (ARBs), are widely used among patients with cardiovascular or kidney disease.^{2–4} The management of RASi in patients with AKI remains a topic of ongoing research.⁵ Several studies have reported that ACEIs or ARBs therapy after discharge is associated with improved outcomes in hospitalized patients with AKI.^{6–8} However, there are no published studies that investigate the survival benefit or harm of RASi treatment during an episode of AKI. It remains inconclusive whether RASi should be stopped or continuously used when AKI was detected in those patients who had been receiving RASi for a long time. This uncertainty represents a common dilemma in clinical practice, where evidence to guide decision making is limited.

To address this knowledge gap, we conducted a multi-database cohort study using sequential target trial emulation framework. Adult patients with hospital-acquired AKI who had been receiving RASi treatment for at least 90 days were included to compare the clinical outcomes between those with and without early discontinuation of RASi during an episode of AKI.

Methods

Target Trial Protocol Overview

We emulated a sequential target trial^{9–11} on a daily basis to evaluate the association between discontinuation of RASi within 2 days after AKI detection with all-cause mortality, compared with continuation of RASi, in a multidatabase cohort study. The elements of the target and emulation trial are presented in [Supplemental Table 1](#). We categorized patients into a discontinuation group and a continuation group based on their RASi prescription records within 2 days after the onset of AKI, adhering to the intention-to-treat (ITT) principle. Emulation of sequential trials on a time window of 24 hours is a valid and efficient procedure to mitigate selection and immortal time bias, which was successfully applied in previous studies comparing treatment versus nontreatment.^{12–14}

Data were extracted from the China Renal Data System (CRDS) and Medical Information Mart for Intensive Care IV (MIMIC-IV version 2.0) databases. CRDS is a nationwide database containing clinical data of more than 10 million inpatients admitted between January 2000 and December 2022 across 28 regional medical centers in China. This large and high-quality database has served as a valuable resource for multiple research studies.^{15–17} The database includes demographic data, diagnostic information, operation and procedure records, prescription data, laboratory measurements, and imaging

and histopathologic reports during each inpatient and outpatient visit. MIMIC-IV is a large, deidentified dataset of patients admitted to the emergency department or intensive care units (ICUs) at the Beth Israel Deaconess Medical Center in Boston, Massachusetts, covering data for over 65,000 patients admitted to the ICU and over 200,000 patients admitted to the emergency department. MIMIC-IV contains comprehensive information on laboratory measurements, medications administered, vital signs documented, and so on.^{18,19}

Eligibility Criteria

Eligible candidates included adult patients who had experienced hospital-acquired AKI and who had been using RASi for at least 90 days prior to the hospital-acquired AKI. We excluded patients with concomitant use of ACEIs and ARBs, a baseline eGFR <30 ml/min per 1.73 m², or vasopressor use. Due to the design of sequential target trial emulation, only the first 2 days after AKI were considered in the study to generate person-trials for analyses (see details below in the Methods section). The eligibility was reassessed repeatedly within the 2-day enrollment period on a daily basis of each “trial” since the onset date of AKI.

AKI was defined according to the creatinine criteria of Kidney Disease Improving Global Outcomes²⁰: a ≥ 0.3 mg/dl increase in serum creatinine concentration within 48 hours or a $\geq 50\%$ increase over baseline serum creatinine within 7 days. AKI that occurred after 2 days following admission was considered as hospital-acquired AKI.²¹ Patients who developed AKI within the first 2 days of hospitalization were classified as having community-acquired AKI and were therefore not eligible.

Treatment Strategies and Treatment Assignment

Treatment strategies included discontinuation or continuation of RASi within 2 days after the detection of hospital-acquired AKI. To address immortal time bias, we employed a sequential target trial emulation on a 24-hour time interval, starting from the day of AKI detection and covering the first 2 days thereafter. On the first day after AKI detection, eligible patients were classified into either the discontinuation or continuation group ([Supplemental Figure 1](#)). Patients were assigned to the discontinuation group if there was no record of RASi administration during that time window (*i.e.*, the first day after AKI onset). Assessment of RASi use was based on electronic medical records (only in the CRDS database) and prescription information. Following the same principle, we replicated this process for the second 24-hour interval after AKI detection. Each patient who met the eligibility criteria contributed one person-trial per 24-hour interval and could contribute up to two person-trials in total. Importantly, once a patient was assigned to the discontinuation group, they were no longer eligible to be included in the continuation group in subsequent intervals, regardless of whether they restarted RASi therapy later. In other words, if a patient was classified into the discontinuation group on day 1, they could not contribute a person-trial on day 2. For person-trials in the first 24-hour interval, the index date was defined as the

day following AKI detection. For person-trials in the second 24-hour interval, the index date was defined as the second day following AKI detection.

Baseline serum creatinine was defined as the mean of all available serum creatinine levels from 30 days prior to admission to the first in-hospital serum creatinine test (for patients without serum creatinine measurement prior to the hospitalization, only in-hospital serum creatinine was used).²² All patients were linked to the corresponding system, hence, out-of-hospital mortality was available.

Outcomes and Follow-Up

The primary outcomes were 30- and 180-day all-cause mortality after hospital-acquired AKI. The secondary outcome was AKI recovery, defined as a reduction in serum creatinine from the peak value by at least 50% of the difference between peak and baseline creatinine, along with resolution of AKI status—indicated by a creatinine level $<150\%$ of baseline or $<\text{baseline}+0.3$ mg/dL.^{21,23} The cause and the date of death in the CRDS database were obtained by the national electronic cause-of-death reporting system from Chinese Centers for Disease Control and Prevention.²⁴ Date of death in the MIMIC-IV database was extracted from two sources: the hospital information system and the Massachusetts State Registry of Vital Records and Statistics. Individual patient records from Medical Information Mart for Intensive Care were matched to the vital records using a custom algorithm based on identifiers including name, social security number, and date of birth. As a result of the linkage, out-of-hospital mortality was available for MIMIC-IV patients up to 1 year post-hospital discharge.²⁵ The follow-up period began on the day the treatment strategy was assigned and ended at death or 1 year.

Clinical Variables

Baseline characteristics included age, sex, body mass index, stage of AKI, total volume of 24-hour intravenous fluids, 24-hour urine output (only for the MIMIC-IV database), comorbidities (hypertension, diabetes, cancer, heart failure, stroke, and coronary artery disease), comedication (loop diuretics, statins, calcium channel blockers, and nonsteroidal anti-inflammatory drugs, β -blocker, proton pump inhibitors, aminoglycoside, and hypoglycemic agent), and laboratory measurements (serum creatinine before AKI, hemoglobin, serum albumin, serum potassium). Comorbidities were determined based on the diagnostic codes using International Classification of Diseases, 10th Revision codes (CRDS database) and International Classification of Diseases, 9th Revision codes (MIMIC-IV database) before the index date. The Charlson's comorbidity index was calculated at hospital admission to provide an objective metric of comorbidity burden.²⁶ Comedications were defined as any prescription of the specified drugs within 1 year before the index date. Baseline laboratory measurements were defined as the most recent value obtained within 90 days prior to or on the index date. The eGFR was calculated using the serum creatinine-based CKD Epidemiology Collaboration equation.²⁷ Description of definition, source, and use of the variables is provided in [Supplemental Tables 2 and 3](#). A directed

acyclic graph was applied to identify the confounding factors ([Supplemental Figure 2](#)).

Statistical Analysis

The target trial causal contrasts of interest was the marginal ITT effect of RASi discontinuation on the cumulative incidence of the all-cause mortality compared with continuation. This marginal effect was defined as the difference between the cumulative incidences (risk differences) when all individuals in the target trial population were assigned to either discontinue or continue RASi treatment at baseline. To flexibly model time-varying covariates and nonproportional hazards in survival analysis, we applied pooled logistic regression to estimate discrete-time hazard, treating person-trials from each day as separate observations. The model included both fixed baseline variables (e.g., sex and baseline diagnoses) and time-varying variables (e.g., laboratory measurements and AKI stage). To adjust for dependent censoring, we applied stabilized inverse probability of censoring weights (IPCWs). Specifically, the stabilized IPCW were constructed as the product of conditional probabilities of remaining uncensored at each follow-up interval. The numerator of the stabilized IPCW represented the probability of being uncensored at baseline, modeled using baseline covariates. The denominator represented the conditional probability of remaining uncensored during follow-up, modeled using time-varying covariates and treatment history. Finally, a pooled regression model combining person-trials on day 1 and day 2 was used to estimate 30- and 180-day cumulative incidences of death, along with the corresponding risk differences between the two treatment strategies in the marginal structural model. The pooled logistic regression model is an approximation to a proportional hazards model when time is split into discrete intervals (e.g., by person-days or person-months).²⁸ The statistical method used in our study accommodates dynamic treatment effects and time-dependent confounding, while using robust sandwich variance estimators to account for intra-individual correlation, aligning with the causal inference framework for observational data.

The last observation carried forward method was used to fill the missing covariates longitudinally. For each covariate with missing data at baseline, we applied median imputation and introduced an indicator variable for the missingness.

Additionally, we performed the per-protocol analyses of RASi discontinuation with mortality. In per-protocol analyses, follow-up data are artificially censored upon treatment switching/deviation (reinitiating a RASi for discontinuation group or stopping a RASi for continuation group) during the follow-up, and the inverse probability of treatment weights (IPTWs) were used to adjust for selection bias introduced by treatment switching. The covariate sets used for IPCW, IPTW, and the pooled logistic regression were identical, encompassing both baseline and time-varying covariates like vital sign (BP), comedications, and laboratory measurement.

The treatment effects were estimated and compared across subgroups stratified by age (<70 or ≥ 70 years), sex, AKI stage, baseline eGFR (≥ 45 or <45 mL/min per

1.73 m²), Charlson's comorbidity index (>6 or ≤6), baseline serum potassium (>5.5 or ≤5.5 mEq/L), ICU admission, presence of diabetes, hypertension, and cancer, and use of loop diuretic.

Seven sensitivity analyses were conducted. First, in the pooled logistic regression model, we further adjusted for the degree of AKI recovery (defined as the ratio of eGFR during follow-up to baseline eGFR) and included it as a time-varying covariate in the pooled logistic regression. Second, we separated patients into those using ACEIs and those using ARBs, and separately investigated the association between discontinuation versus continuation of study drug and mortality within each subgroup. Third, we excluded patients with hyperkalemia (baseline serum potassium >5.5 mEq/L) at the time of AKI detection. Fourth, we further excluded those who died within 5 days after AKI detection to preclude the reversal causality. Fifth, we explored different time intervals for treatment assignment, redefining the windows as ≤1, ≤2 (the primary analysis time window), and ≤7 days, and reanalyzed the association between discontinuation versus continuation of RASi under each updated treatment strategy. Sixth, we performed a multiple imputation under a missing-at-random framework by chained equations to evaluate the robustness of the primary findings. Seventh, we performed a 1:2 propensity score-matched analysis (caliper width: 0.25 SDs of logit odds) to balance the baseline covariates among RASi discontinuation and continuation groups. After matching, the identical approach as the primary analysis was applied to the matched population. All statistical analyses were conducted separately in the two independent datasets (CRDS and MIMIC-IV) using R (version 4.1.2).

Results

Study Population and Baseline Characteristics

A total of 27,003 person-trials were identified, each involving patient who had been using RASi for at least 90 days prior to the onset of hospital-acquired AKI (21,680 from the CRDS database and 5323 from the MIMIC-IV database; [Figure 1](#)). The median age of person-trials in the CRDS cohort was 70 (interquartile range, 58–78) years, with 60% being male, whereas in the MIMIC-IV database, the median age was 73 (interquartile range, 63–82) years, with 53% being male ([Table 1](#)). Baseline serum creatinine measurements were

predominantly obtained on the day of hospital admission in both CRDS and the MIMIC-IV databases ([Supplemental Figure 3](#)). Missing data proportions for baseline variables in the RASi discontinuation and continuation groups are summarized in [Supplemental Table 4](#). RASi were discontinued within 2 days after AKI in 2948 (14%) person-trials in the CRDS cohort and 1468 (28%) in the MIMIC-IV cohort. The doses of RASi at the time of discontinuation are detailed in [Supplemental Table 5](#). Person-trials in the discontinuation group exhibited a higher proportion of severe AKI and lower baseline eGFR compared with those in the continuation group in both databases. Comorbidities and comedication usage were generally similar between the discontinuation and continuation groups ([Table 1](#)). In person-trials who discontinued RASi after the AKI, the baseline characteristics of these person-trials for those who discontinued on the first and second day are shown in [Supplemental Table 6](#). The departments in which patients experience hospital-acquired AKI in the two databases are shown in [Supplemental Figure 4](#).

Discontinuation of RASi and Outcomes

Among the 21,680 person-trials in the CRDS database ([Table 2](#)), the cumulative incidence of 30-day all-cause mortality was 4.36% (95% confidence interval [CI], 3.51% to 6.23%) for those who discontinued RASi compared with 5.91% (95% CI, 5.42% to 6.92%) for those who continued, resulting in a significant risk difference of −1.55% (95% CI, −2.43% to −0.55%) between the two groups. At the 180-day follow-up, discontinuation of RASi was associated with −3.12% (95% CI, −4.94% to −1.03%) lower risk of all-cause mortality compared with the continuation group. In the MIMIC-IV database ([Table 2](#)), the cumulative incidences of all-cause mortality within 30 and 180 days were higher than that in the CRDS database. However, the risk difference at 30 and 180 days between person-trials with RASi discontinuation and continuation was attenuated to −0.79% (95% CI, −1.61% to −0.12%) and −1.98% (95% CI, −3.80% to −0.28%), respectively. [Figure 2](#) illustrates the ITT 180-day adjusted cumulative incidence curves for the discontinuation versus continuation groups. The risk difference associated with RASi discontinuation on all-cause mortality was consistent across subgroups stratified by age, sex, baseline eGFR, AKI stage, comorbidity, and loop diuretics, and this consistency was observed in both databases ([Figure 3](#)). Notably, the beneficial

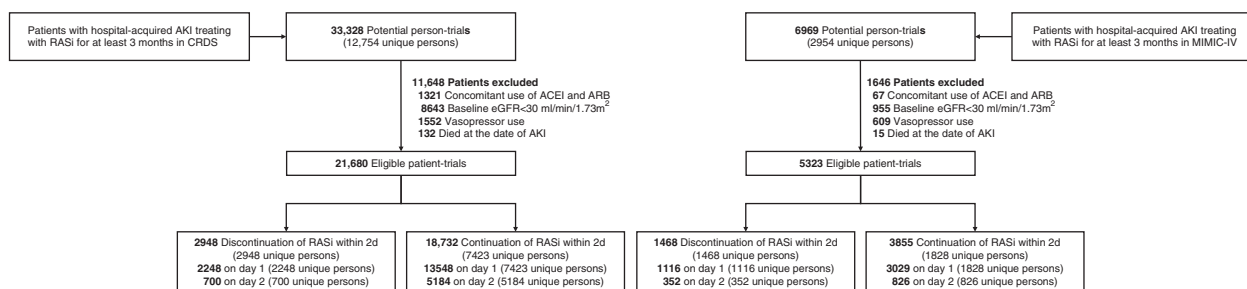


Figure 1. Flow chart of study population. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blockers; CRDS, China Renal Data System; MIMIC-IV, Medical Information Mart for Intensive Care IV; RASi, renin-angiotensin system inhibitors.

Table 1. Baseline characteristics of person-trials with hospital-acquired AKI in the emulated target trial

Characteristics	CRDS			MIMIC-IV		
	Total (21,680 Person-Trials)	Discontinuation of RASi (2948 Person-Trials)	Continuation of RASi (18,732 Person-Trials)	Total (5323 Person-Trials)	Discontinuation of RASi (1468 Person-Trials)	Continuation of RASi (3855 Person-Trials)
Age, yr, median (IQR)	70 (58–78)	70 (59–79)	70 (58–78)	73 (63–82)	73 (63–83)	72 (63–82)
Male, No. (%)	12,972 (60)	1793 (61)	11,179 (60)	2825 (53)	810 (55)	2015 (52)
BMI, kg/m ² , median (IQR)	21.9 (21.0–26.0)	22.0 (21.0–26.0)	22.1 (21.0–26.0)	29.5 (25.6–34.4)	29.1 (25.2–33.8)	29.6 (25.6–34.9)
AKI stage						
1	17,240 (80)	2209 (75)	15,031 (80)	4627 (87)	1253 (85)	3374 (88)
2	3020 (14)	495 (17)	2525 (14)	537 (10)	151 (10)	386 (10)
3	1420 (6)	244 (8)	1176 (6)	159 (3)	64 (4)	95 (2)
ICU, No. (%)	5495 (25)	679 (23)	4816 (26)	1400 (26)	390 (27)	1010 (26)
Cardiac operation, No. (%)	781 (4)	137 (5)	644 (3)	36 (0.7)	8 (0.5)	28 (0.7)
RASi use, No. (%)						
ACEI	7296 (34)	988 (33)	6308 (34)	3677 (68)	1027 (69)	2650 (68)
ARB	14,384 (66)	1960 (67)	12,424 (66)	1684 (32)	452 (31)	1232 (32)
Systolic BP, mm Hg, median (IQR)	125 (113–138)	123 (110–136)	125 (113–138)	123 (107–138)	122 (105–136)	123 (108–140)
Diastolic BP, mm Hg, median (IQR)	73 (65–80)	72 (63–80)	73 (65–80)	61 (52–73)	60 (52–71)	61 (52–73)
Volume of 24-h intravenous fluids, ml, median (IQR)	257 (100–700)	435 (100–1100)	250 (100–620)	1000 (300–2070)	1000 (250–1947)	1000 (314–2100)
24-h urine output, ml, median (IQR)	—	—	—	1300 (720–2100)	1326 (768–2126)	1300 (714–2063)
Baseline laboratory parameters						
Baseline serum creatinine before AKI, mg/dl, median (IQR)	0.9 (0.7–1.2)	1.0 (0.7–1.3)	0.9 (0.7–1.2)	1.0 (0.8–1.3)	1.1 (0.8–1.3)	1.0 (0.7–1.2)
Baseline eGFR before AKI, ml/min per 1.73 m ² , median (IQR)	77 (53–96)	72 (50–94)	77 (53–96)	66 (49–88)	61 (46–83)	68 (51–88)
Hemoglobin, g/dl, median (IQR)	11.7 (9.9–13.4)	11.6 (9.7–13.3)	11.8 (10.0–13.4)	10.7 (9.4–12.0)	10.6 (9.1–12.1)	10.7 (9.5–12.0)
Serum albumin, g/dl, median (IQR)	3.5 (3.1–3.9)	3.5 (3.0–3.9)	3.5 (3.1–3.9)	3.6 (3.2–4.0)	3.6 (3.2–4.0)	3.6 (3.2–4.0)
Serum potassium, mEq/L, median (IQR)	4.0 (3.7–4.4)	4.1 (3.7–4.5)	4.0 (3.7–4.4)	4.2 (3.9–4.5)	4.2 (3.9–4.6)	4.2 (3.9–4.5)
Comorbidity, No. (%)						
Charlson score	5 (4–7)	5 (4–7)	5 (4–7)	6 (4–8)	6 (4–8)	6 (4–8)
Hypertension	14,179 (65)	1950 (66)	12,229 (65)	4751 (89)	1326 (90)	3425 (89)
Diabetes	6734 (31)	947 (32)	5787 (31)	2648 (50)	725 (49)	1923 (50)
Stroke	4909 (23)	648 (22)	4261 (23)	331 (6)	89 (6)	242 (6)
Cancer	2254 (10)	373 (13)	1881 (10)	741 (14)	210 (14)	531 (14)
Heart failure	8882 (41)	1279 (43)	7603 (41)	2566 (48)	702 (48)	1864 (48)
Coronary artery disease	9924 (46)	1443 (49)	8481 (45)	2101 (40)	603 (41)	1498 (39)
Comedications, No. (%)						
Loop diuretics	16,432 (76)	2286 (78)	14,146 (76)	3549 (67)	940 (64)	2609 (68)
Statins	14,495 (67)	2012 (68)	12,483 (67)	4107 (77)	1118 (76)	2989 (78)
CCB	13,640 (63)	1775 (60)	11,865 (63)	3865 (73)	1070 (73)	2795 (72)
β -blocker	14,386 (66)	1937 (66)	12,449 (66)	2453 (46)	663 (45)	1790 (46)
NSAIDs	15,225 (70)	2113 (72)	13,112 (70)	4242 (80)	1188 (81)	3054 (79)

Table 1. Continued

Characteristics	CRDS			MIMIC-IV		
	Total (21,680 Person-Trials)	Discontinuation of RASi (2948 Person-Trials)	Continuation of RASi (18,732 Person-Trials)	Total (5323 Person-Trials)	Discontinuation of RASi (1468 Person-Trials)	Continuation of RASi (3855 Person-Trials)
Proton pump inhibitors	17,568 (81)	2414 (82)	15,154 (81)	4151 (78)	1141 (78)	3010 (78)
Aminoglycoside	1214 (6)	133 (4)	1081 (6)	3024 (57)	860 (59)	2164 (56)
Hypoglycemic agent	11,440 (53)	1590 (54)	9850 (53)	177 (3)	53 (4)	124 (3)

ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; CCB, calcium channel blockers; CRDS, China Renal Data System; ICU, intensive care unit; IQR, interquartile range; MIMIC-IV, Medical Information Mart for Intensive Care IV; NSAID, nonsteroidal anti-inflammatory drug; RASi, renin-angiotensin system inhibitor.

Table 2. Cumulative incidences of all-cause mortality and risk differences among patients with renin-angiotensin system inhibitors discontinuation and continuation

Study Outcome	CRDS		MIMIC-IV	
	Cumulative Incidence (95% CI)	Risk Difference (95% CI)	Cumulative Incidence (95% CI)	Risk Difference (95% CI)
30-d all-cause mortality				
Continuation	5.91% (5.42 to 6.92)	0 (Reference)	6.71% (6.05 to 7.69)	0 (Reference)
Discontinuation	4.36% (3.51 to 6.23)	−1.55% (−2.43 to −0.55)	5.91% (5.27 to 7.03)	−0.79% (−1.61 to −0.12)
180-d all-cause mortality				
Continuation	13.02% (12.41 to 14.40)	0 (Reference)	19.12% (17.91 to 20.78)	0 (Reference)
Discontinuation	9.90% (8.35 to 13.18)	−3.12% (−4.94 to −1.03)	17.14% (15.25 to 19.43)	−1.98% (−3.80 to −0.28)

CI, confidence interval; CRDS, China Renal Data System; MIMIC-IV, Medical Information Mart for Intensive Care IV.

association between discontinuation of RASi and mortality was more prominent in patients with high risk of mortality (e.g., patients with age ≥ 70 years, AKI stage 3, or cancer). In the per-protocol analysis, reinitiation of RASi within hospitalization occurred in 1346 (45.7%) patients in the CRDS database and 609 (41.5%) patients in the MIMIC-IV database. Baseline characteristics among patients with RASi reinitiation are shown in [Supplemental Table 7](#). After applying the IPTW in the pooled logistic regression, the beneficial association of RASi discontinuation remained ([Supplemental Table 8](#)). Regarding secondary outcomes, approximately 50% of patients recovered within 7 days following AKI, with the estimated risk differences for AKI recovery associated with the discontinuation of RASi being 8.92% (95% CI, 5.17% to 13.13%) in the CRDS database and 6.51% (95% CI, 3.02% to 9.60%) in the MIMIC-IV database, respectively ([Supplemental Table 9](#)).

Sensitivity Analysis

The association between ACRIs/ARBs discontinuation and mortality remained consistent after adjusting for AKI recovery ([Supplemental Table 10](#)), when stratifying by ACEI and ARB users ([Supplemental Table 11](#)), after excluding patients with AKI accompanied by hyperkalemia ([Supplemental Table 12](#)), and after excluding patients who died within 5 days after AKI detection ([Supplemental Table 13](#)), and after multiple imputation ([Supplemental Table 14](#)) in both the CRDS and MIMIC-IV databases. After matching, the baseline covariates were well-balanced between the RASi discontinuation and continuation groups ([Supplemental Table 15](#)). Consistent with the primary analysis, these sensitivity analyses demonstrated that RASi discontinuation was significantly associated with lower risk of mortality ([Supplemental Table 16](#)). In the sensitivity analysis using different time intervals for defining RASi discontinuation ([Supplemental Figure 5](#)), we found that person-trials with RASi discontinuation within the first 2 days (including ≤ 1 and ≤ 2 days) were associated with lower risk of mortality compared with continued use of RASi. As the time interval for discontinuation extended beyond 2 days, the risk differences between the discontinuation and continuation groups diminished.

Discussion

In this multidatabase cohort study, we emulated a sequence of hypothetical randomized trials comparing discontinuation versus continuation of RASi in adults with AKI during hospitalization. Our study, consistent across both the CRDS and MIMIC-IV databases, found that stopping RASi within the first 2 days after the detection of AKI was associated with a significantly lower risk of all-cause mortality at both 30- and 180-days follow-up. The beneficial association of early discontinuation of RASi was also observed across subgroups stratified by clinical variable and multiple sensitivity analyses. To our knowledge, this is the first study investigating the relationship between early discontinuation or continuation of the prior used RASi with all-cause mortality during an episode of AKI.

There is growing evidence regarding the management of RASi in patients with AKI, although most of previous studies focused on the associations between the initiation or resumption of RASi after discharge and subsequent outcomes. Despite the established benefits from the use of RASi after discharge,^{6,7,29–32} evidence remains lacking on whether RASi should be temporarily stopped or continuously used during the episode of AKI. The issue is attracting much attention as many hospitalized patients with AKI are previous RASi users for treating comorbidities, e.g., diabetes, chronic heart failure.³³ The theoretical contribution of RASi to AKI may lead to the practice of discontinuing RASi during acute illness such as AKI.³⁴ However, there is a paucity of evidence to support this experience to improve clinically relevant outcomes.^{35–37} Similarly, the Kidney Disease Improving Global Outcomes guidelines provide a practice point (consensus-based recommendations that reflect expert judgment but are not graded), suggesting that temporarily withholding RASi in the peri-operative or acute settings may help prevent AKI.³⁸ However, the supporting evidence is inconsistent, influenced by the small sample sizes and limited scope of existing studies.

With a rigorous study design, our results showed that early discontinuation of RASi during an episode of AKI was associated with a lower risk of all-cause mortality across 27,003 person-trials from two independent databases. When considered alongside previous studies demonstrating lower risk of death with RASi use after AKI

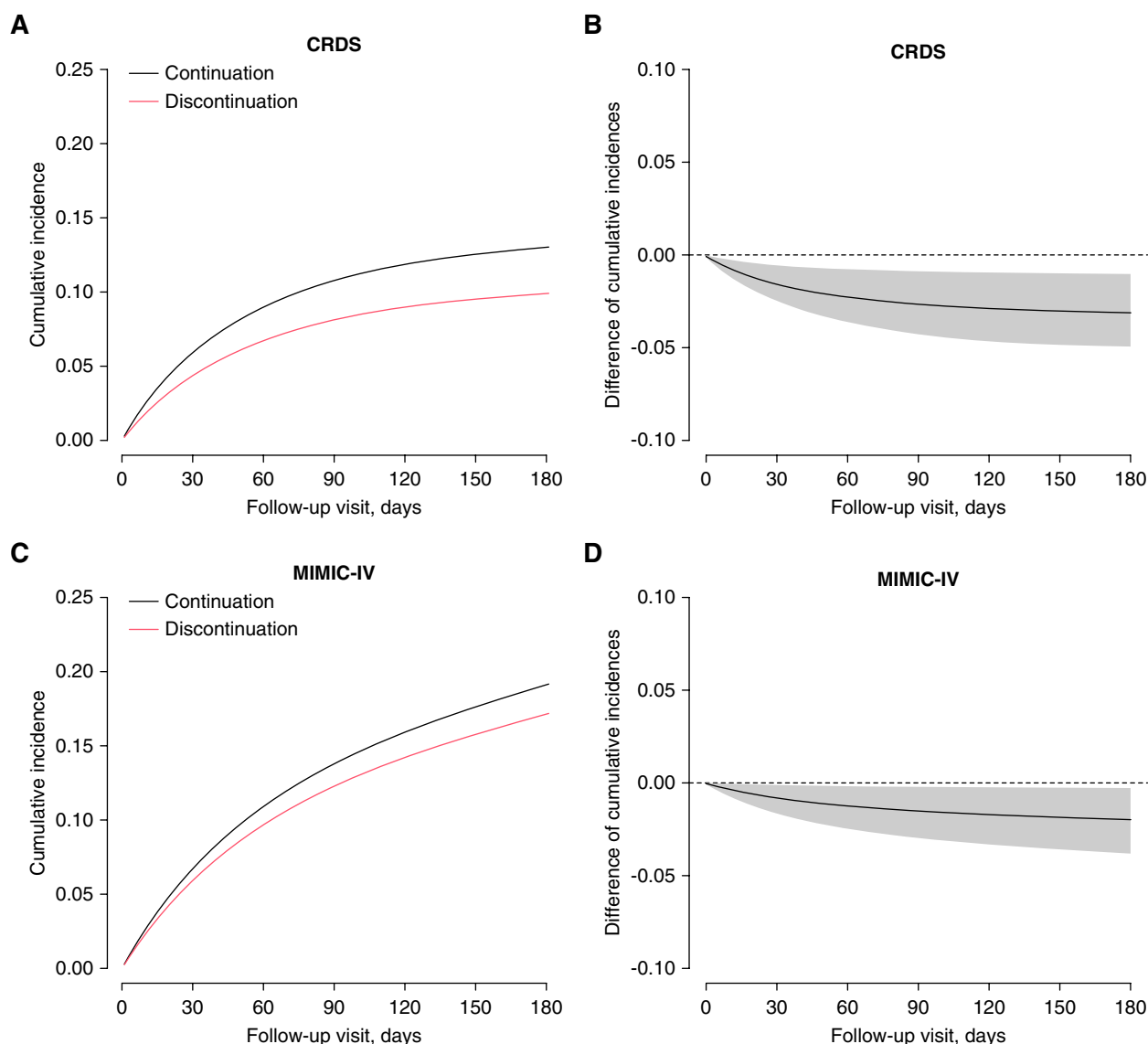


Figure 2. Cumulative incidences and risk differences of patients with RASi discontinuation and continuation within 180 days. (A) Cumulative incidence of all-cause mortality in the CRDS database. (B) Risk difference of patients with RASi discontinuation and continuation in the CRDS database. (C) Cumulative incidence of all-cause mortality in the MIMIC-IV database. (D) Risk difference of patients with RASi discontinuation and continuation in the MIMIC-IV database. Figure 2 can be viewed in color online at www.jasn.org.

discharge,^{6,7,29–32} an important and unresolved question emerges: What is the optimal timing to initiate or restart RASi following AKI? For example, a retrospective cohort study from the Alberta Kidney Disease Network found that patients who initiated RASi within 90 days post-discharge had a higher risk of mortality compared with those who started after 90 days (hazard ratio, 1.15; 95% CI, 1.03 to 1.28).⁶ While our findings suggest a potential strategy for managing RASi during AKI, further prospective studies are needed to validate these results and determine the safest and most effective timing for RASi reinitiation.

It is worth noting that our study reported absolute risk differences rather than risk ratios, providing a transparent depiction of the magnitude of the mortality benefit associated with RASi discontinuation within 2 days after AKI. Although the observed absolute reduction in mortality risk was modest—approximately 1%–3%—this finding

should be interpreted in the context of the relatively low baseline mortality in our study population. Specifically, the 30-day mortality risk among patients with AKI was approximately 5%–6%, and the 180-day mortality ranged from 10% to 19%. Against this background, even a 1%–3% absolute risk reduction represents a meaningful proportional improvement in patient outcomes, particularly given the high prevalence of AKI and the widespread use of RASi in hospitalized settings. Therefore, despite the modest numerical value, the observed benefit remains clinically relevant.

The mechanism by which early discontinuation of RASi after AKI is associated with a lower risk of death remains unclear. One potential explanation, hinted by preliminary clinical studies, is that patients using RASi may observe a slight increase in GFR after discontinuation of such drugs, possibly due to the impact of RASi on

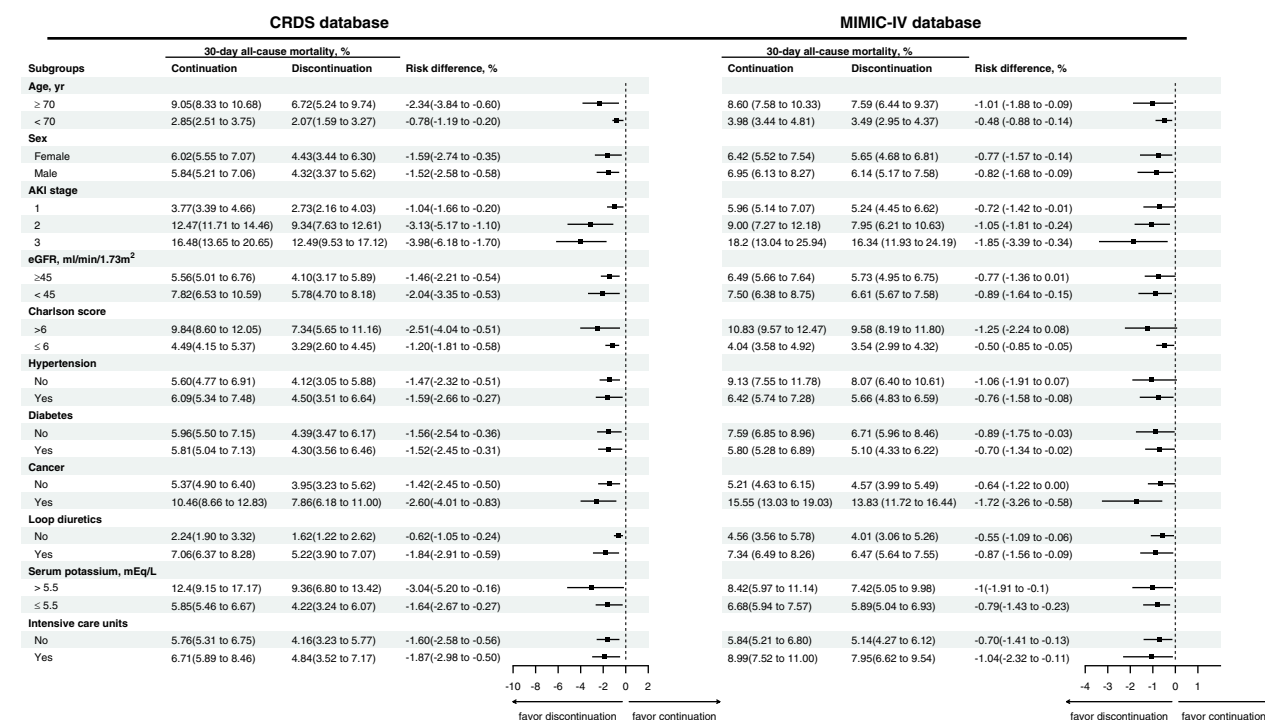


Figure 3. Subgroup analysis for the 30-day risk difference of patients with RASi discontinuation and continuation. Figure 3 can be viewed in color online at www.jasn.org.

glomerular hemodynamics.³⁹ This response might be beneficial for alleviating kidney stress during AKI. Besides, the lower risk of death observed with RASi discontinuation in our study may be secondary to the protection of kidney outcomes. Previous studies have confirmed that the duration and severity of AKI are important risk factors for AKI mortality.^{21,40–42} Supporting this notion, the rate of AKI recovery before discharge in the discontinuation group was significantly higher than that in the continuation group in both the CRDS and MIMIC-IV databases.

Our study has several strengths. First, it utilizes two large real-world databases, allowing for a robust analysis based on a substantial sample size that covers a wide spectrum of diseases, lending high credibility to our findings. The consistency observed across both independent databases further strengthens the robustness of our results. Additionally, the comprehensive patient-level data ensures the ability to adjust for important confounders such as the potential reason for drug discontinuation (*e.g.*, hyperkalemia, severe AKI, ICU admission, and mechanical ventilation), baseline comorbidities, and other drug use. Second, misclassification of assignment to treatment strategies based on treatment received after eligibility results in immortal time when the treatment strategies are not distinguishable at the start of follow-up in observational study. We employed a sequential target trial emulation framework to align eligibility with assignment to treatment strategies at time zero, which prevents the selection and misclassification that can lead to immortal time. Furthermore, by using mortality as a hard end point, we improved the study's relevance to clinical outcomes.

Nonetheless, there are some limitations to our analysis. First, the observational nature of our study precludes the ability to confirm the causality, even though the results are consistent in subgroups and multiple sensitive analyses. Second, clinicians were more likely to stop RASi in sicker patients, such as those with hypovolemic, hyperkalemia, and mechanical ventilation. Although we have adjusted these common causes of RASi discontinuation, the residual unmeasured confounding factors related to the RASi discontinuation may persist as there is no consensus for RASi discontinuation. Third, we are not able to determine the etiology of AKI, and whether there was any difference in the effect of RASi among different types of AKI is unclear. Fourth, only a minority of patients had creatinine measurements in the 30 days prior to admission and, thus, most patients had their baseline creatinine defined at hospital admission.

In conclusion, early discontinuation of RASi during an episode of AKI is associated with a lower all-cause mortality in patients previously treated with RASi. Further research is needed to confirm these findings and assess their applicability in clinical practice with confidence.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JASN/F285>.

Author Contributions

Conceptualization: Sheng Nie, Xin Xu.

Data curation: Shan Chen, Sheng Nie, Mingzhen Pang, Yinfang Sun.

Formal analysis: Ruixuan Chen, Yanqin Li, Shiyu Zhou.

Funding acquisition: Fan Fan Hou.

Investigation: Yanqin Li, Mingzhen Pang, Yinfang Sun.

Methodology: Yanqin Li, Mingzhen Pang, Yinfang Sun, Shiyu Zhou.

Project administration: Sheng Nie, Licong Su.

Software: Ruixuan Chen, Shan Chen, Fan Luo, Licong Su, Shiyu Zhou.

Supervision: Fan Fan Hou, Licong Su, Xin Xu.

Validation: Fan Fan Hou, Xian Shao.

Visualization: Xian Shao, Shiyu Zhou.

Writing – original draft: Ruixuan Chen, Sheng Nie.

Writing – review & editing: Fan Fan Hou, Xin Xu.

Funding

X. Xu: National Key R&D Program of China (2021YFC2500200 and 2021YFC2500204). F.F. Hou: Key Technologies R&D Program of Guangdong Province (2023B1111030004), Guangdong Provincial Clinical Research (2020B111170013), Program of Introducing Talents of Discipline to Universities (D18005), and National Natural Science Foundation of China (82030022). S. Nie: National Natural Science Foundation of China (81900626).

Data Availability Statements

Partial restrictions to the data and/or materials apply. The data that support the findings of this study are available from the National Clinical Research Center for Kidney Disease and CRDS by emailing xux007@163.com or niesheng0202@126.com.

Supplemental Material

This article contains the following supplemental material online at <http://links.lww.com/JSN/F286>.

Supplemental Table 1. Specification and emulation of target trial.

Supplemental Table 2. Variable definitions, sources, and use in analyses.

Supplemental Table 3. Definitions of comorbidities and comedications.

Supplemental Table 4. Missing data in baseline characteristics between discontinuation and continuation of RASi.

Supplemental Table 5. Prescription doses of RASi at discontinuation.

Supplemental Table 6. The baseline characteristics for person-trials with RASi discontinuation at the first 2 days.

Supplemental Table 7. The baseline characteristics for person-trials with and without reinitiated RASi after discontinuation.

Supplemental Table 8. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and continuation in per-protocol analysis.

Supplemental Table 9. Cumulative incidences of AKI recovery and risk differences among patients with RASi discontinuation and continuation.

Supplemental Table 10. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and continuation in sensitivity analysis after adjusting the recovery of AKI.

Supplemental Table 11. Cumulative incidences of outcomes and risk differences among patients with ACEI and ARB discontinuation and continuation.

Supplemental Table 12. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and continuation in sensitivity analysis after excluding patients with hyperkalemia at AKI detection.

Supplemental Table 13. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and

continuation in sensitivity analysis after excluding patients who died within 5 days after AKI.

Supplemental Table 14. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and continuation after multiple imputation.

Supplemental Table 15. Baseline characteristics of person-trials with hospital-acquired AKI after propensity score matching.

Supplemental Table 16. Cumulative incidences of outcomes and risk differences among patients with RASi discontinuation and continuation after propensity score matching.

Supplemental Figure 1. Schematic illustration of emulated target trial design.

Supplemental Figure 2. Directed acyclic graph.

Supplemental Figure 3. Distribution of serum creatinine testing times relative to admission.

Supplemental Figure 4. Departments of patients with RASi discontinuation and continuation in the two databases.

Supplemental Figure 5. Cumulative incidences of 30-day mortality and risk differences among patients with RASi discontinuation and continuation at different time periods.

References

- Hoste EAJ, Kellum JA, Selby NM, et al. Global epidemiology and outcomes of acute kidney injury. *Nat Rev Nephrol*. 2018; 14(10):607–625. doi:10.1038/s41581-018-0052-0
- Bangalore S, Fakheri R, Wandel S, Toklu B, Wandel J, Messerli FH. Renin angiotensin system inhibitors for patients with stable coronary artery disease without heart failure: systematic review and meta-analysis of randomized trials. *BMJ*. 2017;356:j4. doi:10.1136/bmj.j4.
- Evans M, Carrero JJ, Szummer K, et al. Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers in myocardial infarction patients with renal dysfunction. *J Am Coll Cardiol*. 2016;67(14):1687–1697. doi:10.1016/j.jacc.2016.01.050
- Kidney Disease Improving Global Outcomes (KDIGO) Blood Pressure Work Group. KDIGO 2021 clinical practice guideline for the management of blood pressure in chronic kidney disease. *Kidney Int*. 2021;99(3S):S1–S87. doi:10.1016/j.kint.2020.11.003
- Menez S, Parikh CR. Renin-angiotensin system blockade after acute kidney injury: the plot thickens. *Clin J Am Soc Nephrol*. 2020;15(1):2–4. doi:10.2215/CJN.13801119
- Brar S, Ye F, James MT, Hemmelgarn B, Klarenbach S, Pannu N.; Interdisciplinary Chronic Disease Collaboration. Association of angiotensin-converting enzyme inhibitor or angiotensin receptor blocker use with outcomes after acute kidney injury. *JAMA Intern Med*. 2018;178(12):1681–1690. doi:10.1001/jamainternmed.2018.4749
- Siew ED, Parr SK, Abdel-Kader K, et al. Renin-angiotensin aldosterone inhibitor use at hospital discharge among patients with moderate to severe acute kidney injury and its association with recurrent acute kidney injury and mortality. *Kidney Int*. 2021;99(5):1202–1212. doi:10.1016/j.kint.2020.08.022
- Murphy DP, Wolfson J, Reule S, Johansen KL, Ishani A, Drawz PE. Renin-angiotensin-Aldosterone system blockade after AKI with or without recovery among US veterans with diabetic kidney disease. *J Am Soc Nephrol*. 2023;34(10):1721–1732. doi:10.1681/ASN.0000000000000196
- Su L, Rezvani R, Seaman SR, Starr C, Gravestock I. Tri-alEmulation: an R package to emulate target trials for causal analysis of observational time-to-event data. *J Stat Softw*. 2024; 55(2). arXiv:2402.12083[stat.ME]. doi:10.18637/jss.v000.i00
- Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *Am J Epidemiol*. 2016; 183(8):758–764. doi:10.1093/aje/kwv254
- Hubbard RA, Gatsonis CA, Hogan JW, Hunter DJ, Normand SLT, Troxel AB. “Target trial emulation” for observational studies – potential and pitfalls. *New Engl J Med*. 2024;391(21): 1975–1977. doi:10.1056/NEJMp2407586
- Al-Samkari H, Gupta S, Leaf RK, et al. Thrombosis, bleeding, and the observational effect of early therapeutic anticoagulation

- on survival in critically ill patients with COVID-19. *Ann Intern Med*. 2021;174(5):622–632. doi:[10.7326/m20-6739](https://doi.org/10.7326/m20-6739)
13. Yen FS, Hwu CM, Liu JS, Wu YL, Chong K, Hsu CC. Sodium-glucose cotransporter-2 inhibitors and the risk for dialysis and cardiovascular disease in patients with stage 5 chronic kidney disease. *Ann Intern Med*. 2024;177(6):693–700. doi:[10.7326/m23-1874](https://doi.org/10.7326/m23-1874)
 14. Hernán MA, Sterne JAC, Higgins JPT, Shrier I, Hernández-Díaz S. A structural description of biases that generate immortal time. *Epidemiology*. 2025;36(1):107–114. doi:[10.1097/ede.0000000000001808](https://doi.org/10.1097/ede.0000000000001808)
 15. Zhou S, Su L, Xu R, et al. Statin initiation and risk of incident kidney disease in patients with diabetes. *CMAJ*. 2023;195(21):E729–E738. doi:[10.1503/cmaj.230093](https://doi.org/10.1503/cmaj.230093)
 16. Xu X, Nie S, Xu H, et al. Detecting neonatal AKI by serum cystatin C. *J Am Soc Nephrol*. 2023;34(7):1253–1263. doi:[10.1681/ASN.0000000000000125](https://doi.org/10.1681/ASN.0000000000000125)
 17. Zhou S, Chen R, Liu J, et al. Comparative effectiveness and safety of atorvastatin versus rosuvastatin: a multi-database cohort study. *Ann Intern Med*. 2024;177(12):1641–1651. doi:[10.7326/m24-0178](https://doi.org/10.7326/m24-0178)
 18. Zhou S, Zeng Z, Wei H, Sha T, An S. Early combination of albumin with crystalloids administration might be beneficial for the survival of septic patients: a retrospective analysis from MIMIC-IV database. *Ann Intensive Care*. 2021;11(1):42. doi:[10.1186/s13613-021-00830-8](https://doi.org/10.1186/s13613-021-00830-8)
 19. Johnson A, Bulgarelli L, Pollard T, Horng S, Celi LA, Mark R. *MIMIC-IV (version 2.0)*. PhysioNet; 2022.
 20. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. Section 2: AKI Definition. *Kidney Int Suppl* (2011). 2012;2(1):19–36. doi:[10.1038/kisup.2011.32](https://doi.org/10.1038/kisup.2011.32)
 21. Xu X, Nie S, Liu Z, et al. Epidemiology and clinical correlates of AKI in Chinese hospitalized adults. *Clin J Am Soc Nephrol*. 2015;10(9):1510–1518. doi:[10.2215/CJN.02140215](https://doi.org/10.2215/CJN.02140215)
 22. Su L, Li Y, Xu R, et al. Association of ibuprofen prescription with acute kidney injury among hospitalized children in China. *JAMA Netw Open*. 2021;4(3):e210775. doi:[10.1001/jamanetworkopen.2021.0775](https://doi.org/10.1001/jamanetworkopen.2021.0775)
 23. Kellum JA, Sileanu FE, Bihorac A, Hoste EA, Chawla LS. Recovery after acute kidney injury. *Am J Respir Crit Care Med*. 2017;195(6):784–791. doi:[10.1164/rccm.201604-0799OC](https://doi.org/10.1164/rccm.201604-0799OC)
 24. Zhou M, Wang H, Zeng X, et al. Mortality, morbidity, and risk factors in China and its provinces, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet*. 2019;394(10204):1145–1158. doi:[10.1016/s0140-6736\(19\)30427-1](https://doi.org/10.1016/s0140-6736(19)30427-1)
 25. Lee H-C, Park Y, Yoon SB, Yang SM, Park D, Jung CW. VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. *Sci Data*. 2022;9(1):279. doi:[10.1038/s41597-022-01411-5](https://doi.org/10.1038/s41597-022-01411-5)
 26. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J Chronic Dis*. 1987;40(5):373–383. doi:[10.1016/0021-9681\(87\)90171-8](https://doi.org/10.1016/0021-9681(87)90171-8)
 27. Delgado C, Baweja M, Crews DC, et al. A unifying approach for GFR estimation: recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease. *J Am Soc Nephrol*. 2021;32(12):2994–3015. doi:[10.1681/ASN.2021070988](https://doi.org/10.1681/ASN.2021070988)
 28. Hernán MA, Brumback B, Robins JM. Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men. *Epidemiology*. 2000;11(5):561–570. doi:[10.1097/00001648-200009000-00012](https://doi.org/10.1097/00001648-200009000-00012)
 29. Yang CY, Liu JS, Tseng WC, et al. Effect of renin-angiotensin-aldosterone system blockade on long-term outcomes in post-acute kidney injury patients with hypertension. *Crit Care Med*. 2020;48(12):e1185–e1193. doi:[10.1097/ccm.0000000000004588](https://doi.org/10.1097/ccm.0000000000004588)
 30. Hsu CY, Liu KD, Yang J, et al. Renin-angiotensin system blockade after acute kidney injury (AKI) and risk of recurrent AKI. *Clin J Am Soc Nephrol*. 2020;15(1):26–34. doi:[10.2215/CJN.05800519](https://doi.org/10.2215/CJN.05800519)
 31. Gayat E, Hollinger A, Cariou A, et al. Impact of angiotensin-converting enzyme inhibitors or receptor blockers on post-ICU discharge outcome in patients with acute kidney injury. *Intensive Care Med*. 2018;44(5):598–605. doi:[10.1007/s00134-018-5160-6](https://doi.org/10.1007/s00134-018-5160-6)
 32. Bidulka P, Fu EL, Leyrat C, et al. Stopping renin-angiotensin system blockers after acute kidney injury and risk of adverse outcomes: parallel population-based cohort studies in English and Swedish routine care. *BMC Med*. 2020;18(1):195. doi:[10.1186/s12916-020-01659-x](https://doi.org/10.1186/s12916-020-01659-x)
 33. Mansfield KE, Nitsch D, Smeeth L, Bhaskaran K, Tomlinson LA. Prescription of renin-angiotensin system blockers and risk of acute kidney injury: a population-based cohort study. *BMJ Open*. 2016;6(12):e012690. doi:[10.1136/bmjopen-2016-012690](https://doi.org/10.1136/bmjopen-2016-012690)
 34. Ostermann M, Bellomo R, Burdmann EA, et al. Controversies in acute kidney injury: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Conference. *Kidney Int*. 2020;98(2):294–309. doi:[10.1016/j.kint.2020.04.020](https://doi.org/10.1016/j.kint.2020.04.020)
 35. Duong H, Tesfaye W, Van C, et al. Sick day management in people with chronic kidney disease: a scoping review. *J Nephrol*. 2023;36(5):1293–1306. doi:[10.1007/s40620-022-01497-5](https://doi.org/10.1007/s40620-022-01497-5)
 36. Fink JC, Maguire RM, Blakeman T, et al. Medication holds in CKD during acute volume-depleting illnesses: a randomized controlled trial of a “sick-day” protocol. *Kidney Med*. 2022;4(9):100527. doi:[10.1016/j.xkme.2022.100527](https://doi.org/10.1016/j.xkme.2022.100527)
 37. Watson KE, Dhaliwal K, McMurtry E, et al. Sick Day medication guidance for people with diabetes, kidney disease, or cardiovascular disease: a systematic scoping review. *Kidney Med*. 2022;4(9):100491. doi:[10.1016/j.xkme.2022.100491](https://doi.org/10.1016/j.xkme.2022.100491)
 38. Kidney Disease Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2024;105(4s):S117–S314. doi:[10.1016/j.kint.2023.10.018](https://doi.org/10.1016/j.kint.2023.10.018)
 39. Coca SG, Garg AX, Swaminathan M, et al. Preoperative angiotensin-converting enzyme inhibitors and angiotensin receptor blocker use and acute kidney injury in patients undergoing cardiac surgery. *Nephrol Dial Transplant*. 2013;28(11):2787–2799. doi:[10.1093/ndt/gft405](https://doi.org/10.1093/ndt/gft405)
 40. Wu C, Zhang Y, Nie S, et al. Predicting in-hospital outcomes of patients with acute kidney injury. *Nat Commun*. 2023;14(1):3739. doi:[10.1038/s41467-023-39474-6](https://doi.org/10.1038/s41467-023-39474-6)
 41. Yang L, Xing G, Wang L, et al. Acute kidney injury in China: a cross-sectional survey. *Lancet*. 2015;386(10002):1465–1471. doi:[10.1016/s0140-6736\(15\)00344-x](https://doi.org/10.1016/s0140-6736(15)00344-x)
 42. Chertow GM, Burdick E, Honour M, Bonventre JV, Bates DW. Acute kidney injury, mortality, length of stay, and costs in hospitalized patients. *J Am Soc Nephrol*. 2005;16(11):3365–3370. doi:[10.1681/ASN.2004090740](https://doi.org/10.1681/ASN.2004090740)